

Production probability, transport, and grammatical possibility

Reanalysing the English dative alternation with open data

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Abstract

This paper develops a no-new-data reassessment of Bresnan et al. (2007). The current analyses reproduce the production-choice target from `languageR::dative`, add explicit null and partial-pooling checks, and test whether the resulting conditioning profile transports to the Spoken British National Corpus 2014 (BNC2014) dative dataset. It then adds a compact DAIS acceptability bridge for the same high-frequency verb core. The result is that modern modelling preserves the classic production conclusion; within the shared high-frequency verb set, source-trained structural predictors retain substantial same-row discrimination in BNC2014, while target-native out-of-fold predictions reveal local calibration and residual lexical structure. DAIS preferences show moderate marginal correspondence with the production scores but are systematically more favourable to the double-object alternative, and the production score adds little once the experimental conditions are represented directly. The paper reads these results through projectibility: production probability, acceptability preference, and grammatical possibility are related evidence channels, not one repeated measurement.

Keywords: dative alternation; production probability; transport; projectibility; grammaticality; corpus linguistics; acceptability

1 THE PROBLEM

The English dative alternation is a small problem with a large methodological payoff. Speakers can say “give Kim the book” or “give the book to Kim”, but the choice is uneven. It varies with verb, length, animacy, definiteness, pronominality, discourse status, corpus, and register.

That unevenness is easy to misread. A rare pattern can be lexically restricted, strongly dispreferred, or tied to a discourse context without falling outside the grammar. A production model therefore has two jobs. It has to predict attested choices, and it has to say what kind of grammatical claim those choices support.

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Bresnan et al. (2007) made the case that corpus probabilities can reveal grammatical possibilities that informal judgements miss. This paper asks how that case looks after open-data replication, modern validation, and cross-corpus transport.

The question is one of PROJECTIBILITY: what observing a conditioning profile in one setting licenses us to predict about another (Goodman, 1955). For the dative alternation, the relevant settings are a packaged reanalysis dataset, a later spoken British corpus, and possible acceptability evidence. Keeping those settings apart is the paper’s main discipline.

2 THE ORIGINAL TARGET

Bresnan et al. (2007) is the anchor because it treats the alternation as a grammatical problem rather than as a simple frequency count. The claim is not that whatever occurs often is grammatical. The claim is that observed choices can expose systematic constraints on what speakers treat as available.

That move matters because informal judgements often compress several facts into one verdict. A construction may sound odd because it’s impossible, because it is rare, because the wrong verb is present, because a discourse condition is missing, or because the constructed example strips away the context that would license it. Corpus data can help separate these cases.

The empirical target is production choice. In Bresnan et al.’s setup, the response variable is whether the recipient appears as a noun phrase (NP) in the double-object pattern or as a prepositional phrase (PP) in the prepositional dative. The predictors describe properties of the verb, recipient, theme, discourse status, and corpus source.

This paper preserves that target. It treats production probability as evidence to be interpreted, rather than as a direct measure of grammaticality. The empirical question is whether the production profile survives cleaner modelling and whether it travels beyond the original corpus setting.

3 DATA REPRODUCTION AND REANALYSIS

The analysis starts with `languageR::dative`, distributed with the Comprehensive R Archive Network (CRAN) `languageR` package (Baayen & Shafaei-Bajestan, 2025). The source tarball supplies the packaged `dative.rda` object; the analysis scripts load it and write derived summaries.

The package documentation identifies the sources as Switchboard and the Treebank Wall Street Journal collection. Those sources place the original data in American English: Switchboard is telephone conversation among speakers from the United States, developed in 1990–1991 (Godfrey & Holliman, 1993), and the Treebank Wall Street Journal component is late-1980s American news and financial writing, with one million words of 1989 material and a larger 1987–1989 Wall Street Journal source collection (Charniak et al., 2000; Marcus et al., 1999).

In the packaged dataset, there are 3,263 rows and 15 columns. The response records recipient realization, with 2,414 NP realizations and 849 PP realizations. The dataset contains 75 verb levels and two modalities: 2,360 spoken tokens and 903 written tokens. Speaker identifiers are missing exactly for the written rows.

For the first model sequence, the response stays binary: NP is coded as 1 and PP as 0. The initial split is verb-stratified, with a fixed seed (20260623), so every verb observed in the test set is also represented in training. That split gives 2,637 training rows and 626 test rows.

The denominator is therefore narrow. These rows are already tokens in which a dative recipient was realized as NP or PP. They aren’t all contexts in which a transfer event could have been expressed,

omitted, paraphrased, or packaged in some other construction. The models estimate conditional choice within an attested alternation frame: given that a token enters the dative-alternation sample, which realization appears?

Initial analyses are deliberately plain. A marginal model sets the baseline. Structured models then add modality, length, verb, and the non-verb predictors. Negative-control and simulation checks accompany them: a scrambled-label check and fake-data checks. These tests ask whether the fitted structure beats chance, not whether it proves a grammatical theory.

A second empirical source is the Spoken British National Corpus 2014 (BNC2014) dative dataset (G. Jensen & McGillivray, 2018; G. B. Jensen & McGillivray, 2019). The verified Figshare comma-separated values (CSV) file has 1,839 parsed data rows and 44 columns under a Creative Commons Attribution 4.0 International (CC BY 4.0) licence. The apparent line-count discrepancy is mechanical: the file has 1,839 newline characters but no terminal newline, giving 1,840 logical text lines including the header.

The BNC2014 dative file comes from the Early-Access Subset of the Spoken BNC2014, whose spontaneous conversations were recorded in 2012–2014 and contain speaker and context metadata.

BNC2014 serves as a transport target rather than a second copy of `languageR::dative`. Its outcome field is `Pattern`, where the verb–noun–noun code `VNN` maps to NP realization and the verb–noun–prepositional-phrase code `VNPP` maps to PP realization. The six shared verbs are *give*, *send*, *show*, *sell*, *offer*, and *lend*. All six occur in the packaged `languageR` dataset.

4 MODERN MODELLING UPDATE

The `languageR` result is a robustness result. The central question is whether modern modelling changes Bresnan et al.’s linguistic conclusion or mainly repackages it statistically. The answer, so far, is the latter. The table reports log loss, area under the receiver operating characteristic curve (AUC), and a compact parameter count.

The non-verb `languageR` formula uses modality, semantic class, recipient length, recipient animacy, recipient definiteness, recipient pronominality, theme length, theme animacy, theme definiteness, theme pronominality, and recipient/theme accessibility. The fixed-verb model adds verb as a fixed effect. The hierarchical model uses the same non-verb fixed effects and a varying verb intercept. All are logistic models for NP versus PP realization with R treatment contrasts and no interactions.

Model	Test log loss	Test AUC	Param.
Marginal null	0.554	0.500	1
Non-verb main model	0.271	0.932	18
Fixed-verb full model	0.234	0.951	92
Hierarchical verb-intercept model	0.233	0.952	19

Table 1: Held-out `languageR::dative` results for the main model sequence.

The non-verb model already predicts held-out production choices well. Adding fixed verb effects improves test log loss from 0.271 to 0.234 and raises AUC from 0.932 to 0.951. The improvement confirms that lexical conditioning is not a minor residue after length, animacy, definiteness, pronominality, and accessibility are included.

The hierarchical verb-intercept model reaches almost the same held-out performance with a smaller fixed-effect parameterization. Its test log loss is 0.233 and its AUC is 0.952. The table’s parameter count for this model is the 18 fixed-effect coefficients plus the verb-intercept variance parameter; it is not an effective degrees-of-freedom estimate for the 75 partially pooled verb effects. The estimated verb random-intercept standard deviation is 2.106, so the lexical signal remains large under partial pooling.

The negative-control and simulation checks behave as they should. Scrambled-label fits lose predictive value. Fake-data fits without a verb signal collapse toward singular zero-variance random-effects models, while a fake verb-effect check recovers a nonzero verb standard deviation. These checks make the production result more credible, but they do not change its explanatory level.

The result supports Bresnan et al.’s production-choice conclusion. It does not create a new grammatical conclusion on its own. The same-data analysis shows that the classic profile is reproducible and statistically stable. The paper’s new empirical pressure has to come from transport.

5 GENERALIZATION AND ACCEPTABILITY

The BNC2014 analysis asks whether the production profile travels across corpus design, variety, period, and annotation regime. The transport model is trained on the spoken languageR rows restricted to the six BNC2014 verbs. It is then scored on harmonized BNC2014 rows, with complete-case filtering stated as part of the estimand.

Three targets are kept apart. The languageR models estimate conditional production choice among attested NP and PP dative tokens. The BNC2014 analysis estimates out-of-domain predictive performance over complete-case rows in the six shared high-frequency verbs after harmonization. DAIS estimates controlled double-object preference for constructed alternatives. These numbers can therefore be read together only as different evidence channels, not as repeated measurements of one quantity.

Cross-variety probabilistic grammar of the dative is not new ground. Bresnan and Ford (2010) compared American and Australian English and found that speakers’ probabilistic choice knowledge is largely shared across varieties, while Röthlisberger et al. (2017) found broad stability across nine post-colonial Englishes. The bare finding that dative conditioning largely travels would not, on its own, be new. The contribution here is the strict transport test, the open harmonization record, and the separation of production probability from acceptability preference and grammatical possibility.

The comparison is a bundled out-of-domain test. The source side is American telephone conversation from 1990–1991, restricted to the six shared spoken languageR verbs. The target side is spontaneous British conversation from 2012–2014, under a different annotation scheme. Those six verbs account for 2,201 of 3,263 languageR rows and for 1,564 of 2,360 spoken rows. They are the high-frequency core, not a random sample of the alternation.

The transport models are logistic generalized linear models with logit link. The core formula is

$$\begin{aligned} \text{Pr}(\text{NP}) \sim & \text{Verb} + \text{recipient length} + \text{theme length} \\ & + \text{recipient animacy} + \text{recipient pronominality} \\ & + \text{theme animacy} + \text{theme definiteness} + \text{theme pronominality}. \end{aligned}$$

Lengths are raw word counts, with no centering, scaling, or logging. Factors use R treatment contrasts, with *give* as the verb reference and “no” as the binary reference. There are no interactions in the

transport models. Factor levels and coding are fixed before scoring BNC2014; no BNC2014-specific scaling or imputation enters the headline model. Recipient definiteness is unavailable in BNC2014 and enters only as a labelled regex proxy in a sensitivity check.

Harmonization is lossy. The outcome maps cleanly: VNN becomes NP and VNPP becomes PP. Verb maps cleanly after the six-verb restriction. Length needs more care. The released BNC2014 length fields are character counts: the data dictionary defines RecLen and ThemeLen as the number of characters in the recipient and theme (G. B. Jensen & McGillivray, 2019). Those fields are not comparable to languageR’s token-like lengths, so the scripts derive word counts from the released phrase strings. Rows with empty phrase strings cannot be used in the complete-case transport model.

The core transport model scores the 1,621 BNC2014 rows complete for the harmonized predictors, not all 1,839 released dative rows. The missing rows are outcome-skewed: NP recipients make up 0.819 of the complete rows but only 0.670 of the 218 incomplete rows. The variable-level pattern is also uneven. Recipient-pronominality is missing for 112 rows with NP rate 0.357, while theme-pronominality is missing for 139 rows with NP rate 0.950. The headline comparison is consequently a complete-case estimate over the shared high-frequency core.

A BNC2014 metadata audit makes the domain label concrete. In the released dative rows, Level 1 dialect is “uk” for 1,720 of 1,839 rows and Level 2 dialect is “english” for 1,689. Gender is female-coded for 1,138 rows and male-coded for 699, and the largest age band is 19–29 with 736 rows. Close-family, partner, and very-close-friend conversations contribute 1,171 rows. The complete-case subset has the same broad shape. A single BNC2014 transport score should therefore be read as a corpus-domain diagnostic, not as a clean estimate for British spoken English in general.

The released BNC2014 file also limits the resampling design. The supporting scripts show that an intermediate `main_speaker_id` was derived and selected, but it was dropped before the public 44-column CSV. No stable speaker, text, or conversation identifier is recoverable from the released files. The native BNC2014 benchmark therefore uses repeated verb-stratified row-level cross-validation, not grouped speaker or conversation folds. The comparison below repairs the old sample mismatch by scoring source-trained and native out-of-fold predictions on the same 1,621 rows, but it cannot remove all within-conversation dependence.

Origin	Model family	Log loss	AUC
languageR source	marginal	0.473	0.500
languageR source	verb only	0.456	0.646
languageR source	non-verb	0.296	0.899
languageR source	full core	0.308	0.906
BNC2014 native OOF	marginal	0.473	0.395
BNC2014 native OOF	verb only	0.439	0.538
BNC2014 native OOF	non-verb	0.234	0.920
BNC2014 native OOF	full core	0.227	0.936

Table 2: Same-row BNC2014 complete-case evaluation. OOF = out-of-fold. All rows are the same 1,621 BNC2014 complete-case observations; the native rows use five repeated 10-fold row-level cross-validation.

The source-trained structural profile transports, but not because shared verb base rates do most

of the work. The source full model improves log loss over the source marginal model by 0.165 in a paired row bootstrap ($[-0.198, -0.133]$ for full minus marginal) and raises AUC by 0.406. The source verb-only model gives only a small log-loss gain over marginal (0.456 vs. 0.473), while the source non-verb model reaches 0.296 log loss and 0.899 AUC. Adding the source verb terms to the non-verb model improves ranking slightly but worsens log loss slightly (full minus non-verb = 0.012, bootstrap interval $[-0.006, 0.030]$). In this target corpus, the portable part of the profile is mainly the structural conditioning rather than lexical base-rate ordering.

The target-native model is still better. On the identical rows, the BNC2014 native full model has log loss 0.227 and AUC 0.936. Against the source full model, the paired difference is -0.081 log loss ($[-0.104, -0.059]$) and $+0.030$ AUC ($[0.017, 0.043]$). The native non-verb model is already close to the native full model, which reinforces the same interpretation: the broad conditioning profile is portable, but the target corpus supplies local calibration and residual lexical structure.

Calibration tells the same story. The source full model underpredicts NP realization for *send*, *offer*, and *lend*, and overpredicts it for *sell*. An intercept-only BNC2014 recalibration improves its log loss from 0.308 to 0.301; an intercept-plus-slope recalibration improves it to 0.290, with slope 0.728. That helps, but it still does not reach the target-native full model. The problem is not only prevalence shift.

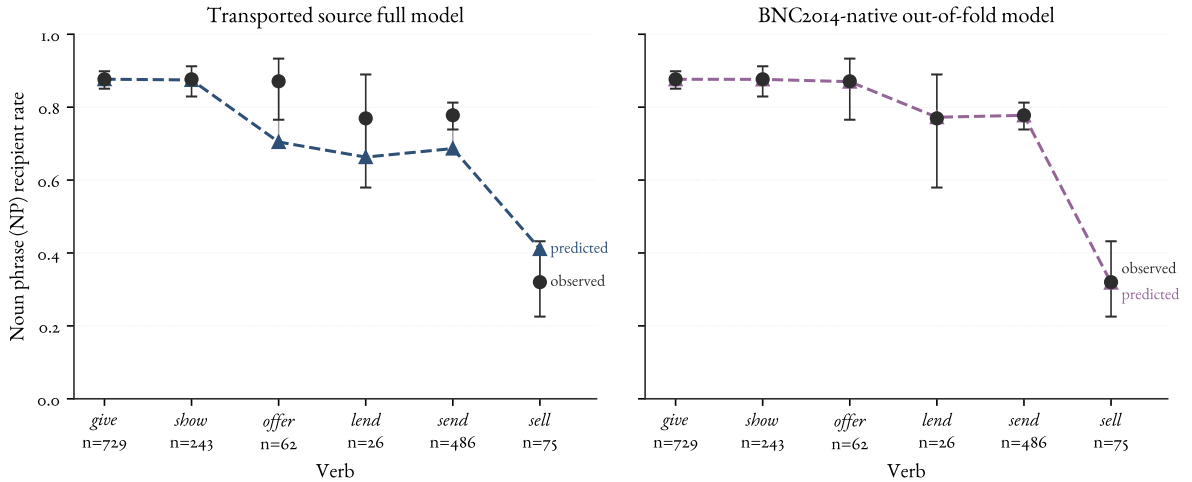


Figure 1: Verb-level calibration for the transported source full model and the BNC2014-native out-of-fold model on the same complete-case rows. Black points and intervals are observed NP recipient rates with Wilson intervals; coloured triangles are mean predictions.

The all-row check is deliberately weaker. Only three common predictors are observed for every BNC2014 row: verb, theme animacy, and theme definiteness. On all 1,839 released rows, the source verb-only model modestly beats the source marginal model (0.469 vs. 0.499 log loss), while the theme variables alone do not recover the complete-case result. This does not replace the headline model; it shows why the headline claim must stay tied to the complete-case harmonized denominator.

Acceptability evidence answers a different question. The Dative Alternation and Information Structure (DAIS) dataset, introduced by Hawkins et al. (2020b), contains 50,000 human judgements over 5,000 dative sentence pairs and systematically varies verb, definiteness, and argument length. The

repository now carries a CC BY 4.0 licence, and the dataset authors have given explicit permission to reuse the judgements. DAIS can therefore enter as a compact bridge (Hawkins et al., 2020a).

The bridge keeps the targets separate. I score only the 150 DAIS items whose verbs overlap the six-verb production core. The same spoken languageR production model that was transported to BNC2014 predicts an NP recipient probability for each constructed DAIS pair; the DAIS item mean gives the corresponding human DO preference on a 0–1 scale. As a bridge analysis, it asks whether a production probability profile and a controlled preference profile point in the same direction.

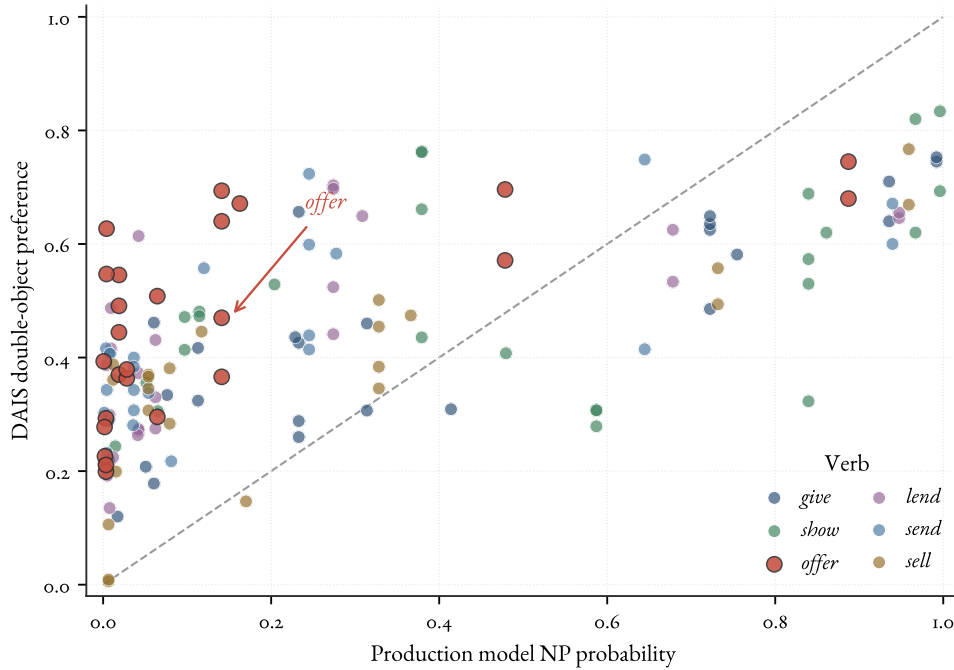


Figure 2: DAIS production/preference bridge for the six shared verbs. Each point is a constructed DAIS item. The horizontal axis is the spoken languageR model’s NP recipient probability; the vertical axis is DAIS DO preference. The diagonal marks equality between the two channels, and the *offer* cluster is highlighted because it shows the clearest channel separation.

The two channels line up moderately, while remaining distinct. With *something* treated as a pronominal theme, Pearson’s r is 0.667 and Spearman’s ρ is 0.685 over the 150 shared-verb items. The mean production NP probability is 0.283, while mean DAIS DO preference is 0.440; the mean paired difference is -0.157 , with bootstrap interval $[-0.198, -0.115]$. Treating *something* as nonpronominal gives the same interpretation ($r = 0.645$, $\rho = 0.675$, mean production probability 0.343).

The item and participant checks show why the raw correlation should not be overread. At the item level, a marginal calibration regression has a positive slope, but once recipient condition, theme type, and verb are represented directly, the production-score slope is essentially zero (0.006, standard error 0.078). At the judgement level, a linear mixed model on 1,525 shared-verb judgements from 923 participants uses fixed effects for production score, recipient condition, theme type, and verb, with crossed random intercepts for participant and constructed item. The fit is non-singular, and the incremental production-score coefficient is 0.017 (standard error 0.076; approximate interval

Verb	Prod. NP prob.	DAIS DO pref.	Prod. - DAIS
<i>give</i>	0.408	0.449	-0.041
<i>lend</i>	0.204	0.426	-0.222
<i>offer</i>	0.150	0.468	-0.319
<i>sell</i>	0.231	0.351	-0.120
<i>send</i>	0.196	0.428	-0.232
<i>show</i>	0.506	0.516	-0.009

Table 3: DAIS bridge for the six shared verbs. The difference column is production probability minus DAIS DO preference.

[−0.132, 0.165]). DAIS preference and production probability therefore show moderate marginal rank correspondence, much of it attributable to shared sensitivity to the manipulated predictors; the production score adds little once those predictors are represented directly.

The useful cases are the divergences. *Offer* is the clearest verb-level one: the production model makes it strongly PP-like, but DAIS participants are much more willing to choose the double-object version. For example, “Juan offered the woman a payment” has predicted NP probability 0.141 but DAIS DO preference 0.694. The point is channel separation: judgement preference can occupy a different region of the evidence space from corpus production probability.

6 GRAMMATICAL POSSIBILITY

The empirical result supports a sharper vocabulary, not a shortcut from frequency to grammar. The languageR analysis shows that a classic production profile is reproducible under explicit validation. The BNC2014 analysis shows that much of that profile travels to the BNC2014 corpus-domain target within the shared high-frequency verb set. Both results concern production choice.

The target is grammaticality, a contested construct that I’ll take as a community-conditioned licensing status for form–value, or form–meaning, relations (Nefdt, 2023; Pullum, 2019). Cognitive and social mechanisms stabilize that status across a speech community. The evidence for that status is graded even when one remains neutral about whether the underlying licensing relation is categorical, multidimensional, or itself graded. A category earns its keep by being projectible for a purpose (Boyd, 1999; Goodman, 1955), and the purpose here is predicting realization in unseen contexts and telling rare-but-grammatical cases apart from ungrammatical ones.

Acceptability preference, raw judgement, corpus frequency, formal generability, and standardness are neighbouring constructs. They can bear on licensing, but none fixes it alone. A corpus token records what a speaker or writer did in a context. A judgement task samples a noisy, role-dependent instrument: how a speaker, participant, or evaluator responds to constructed alternatives under controlled conditions.

The useful distinction is between target and channel. Corpus choices are one channel: they show which form was realized when an alternation token occurred. Judgements are another channel: they provide detector responses calibrated to a speech community and task role. Grammaticality is the target those channels bear on indirectly.

The DAIS bridge shows why the distinction matters. Its shared-verb judgements agree enough with the production model to show shared sensitivity to several dative conditioning factors. They also

diverge enough to show that acceptability preference has a different evidential role from production probability. The *offer* cases are the useful warning: controlled judgements can make a double-object form look more acceptable than a corpus production model trained on attested choices would lead us to expect.

Everything here is conditioned on the opportunity set. This corpus evidence covers attested dative-alternation tokens: cases already annotated as NP or PP recipient realizations. It can support claims about which realization speakers chose inside that frame. A broader corpus design could estimate other opportunities: transfer events expressed by non-dative paraphrase, left implicit in context, or paired with an independent acceptability task. These released rows don't provide that broader denominator.

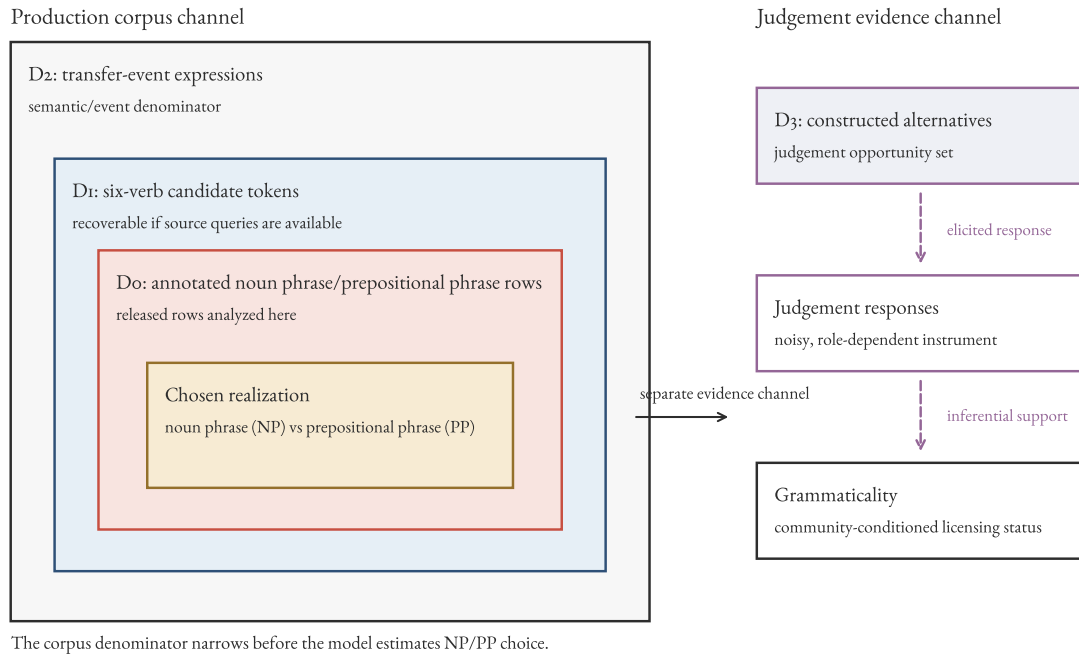


Figure 3: Nested opportunity sets for the dative-alternation evidence. The released corpus rows estimate realization choice inside the annotated noun phrase/prepositional phrase denominator. Broader transfer-event and judgement opportunity sets would answer different questions about grammaticality.

Production data can support claims about grammaticality only through an explicit bridge from attested choices to possible alternatives. Figure 3 marks where that bridge would have to be built. A fitted production profile may be projectible to new tokens; community licensing of a candidate form is a further target that the production profile bears on only through additional assumptions.

The dative alternation makes the bridge visible. A low-probability double-object realization may be grammatical but dispreferred under the observed predictors. A prepositional realization may be likely because the theme is short and the recipient is long. A form may be absent because the corpus lacks the relevant verb, register, or discourse setting.

No amount of additional positive-token data collected under the same D_0 design can, by itself,

identify the grammatical-but-unattested region. More tokens enlarge the attested NP/PP region without changing the opportunity set. That is an observational-identification claim, not a claim that production evidence is useless. A broader denominator, targeted elicited production, novel-item productivity tests, or acceptability judgements can add information about the otherwise empty region.

The dative case makes that boundary measurable. The opportunity set says which cells a production channel can and cannot reach, so the gap stops being a slogan and becomes a property of a stated dataset and design, open to the same kind of modelling the rest of the paper uses.

The current analyses therefore license a cautious claim. Bresnan et al.'s production-conditioning profile is durable and partly transportable within the shared high-frequency core. DAIS shows that related conditioning is visible through judgements, while preference and production can separate. That supports the idea that corpus probabilities can reveal structure that informal judgements miss. It does not make production probability identical to grammaticality. What the durable, transportable profile buys is prediction: it forecasts realization for unseen tokens within its scope of verbs, registers, and varieties, and it marks the low-probability cells where grammaticality has to be settled through another channel.

7 CONCLUSION

The reanalysis has two production results and one acceptability bridge. First, the languageR profile survives baselines, negative controls, simulation checks, and partial pooling. The hierarchical model matches the fixed-verb model with a smaller fixed-effect parameterization, so modern modelling confirms the classic production-choice conclusion.

Second, the profile partly transports. A spoken six-verb model trained on languageR predicts BNC2014 complete-case rows well above a marginal baseline when scored on the same rows as a target-native out-of-fold benchmark. The strongest portable signal is structural rather than verb-only: the source-trained non-verb model nearly matches the source full model by log loss. The BNC2014-native model still performs better. The diagnostic verdict is bounded. Transport survives the null comparison in the high-frequency shared core, but the target corpus is internally structured and the comparison remains too bundled to identify separate variety, period, register, annotation, or missingness effects.

The bridge is DAIS. On the six verbs shared with the production core, human double-object preference tracks the production profile moderately, but the two channels diverge in diagnostically useful places. The item and participant-level checks reinforce that reading: after recipient condition, theme type, and verb are included, the production score has little incremental association with individual judgements. The *offer* cases show why acceptability evidence cannot be replaced by a larger production corpus.

That combination is the paper's centre. The classic result is not overturned. It's made reproducible, tested against real nulls, and moved into a second corpus where it can succeed or fail. The success is strong enough to matter, and the failures are specific enough to guide the next analysis.

The conceptual lesson is the same one that motivates the title. Production probability is evidence about grammatical possibility, not a synonym for it. Open data, transport tests, and the DAIS bridge make that evidence stronger, but they also mark a structural boundary: a production model estimates choice within the attested frame. Additional positive-token data under the same denominator cannot identify the grammatical-but-unattested region by itself. Crossing that boundary needs a broader

opportunity set, an explicit extrapolative model, or another evidence channel.

DATA AND CODE AVAILABILITY

The public working repository is [BrettRey/bresnan-dative-alternation-reanalysis](#). A submission release should archive the exact commit or digital object identifier. The analysis scripts fetch public datasets to temporary locations and validate downloaded files by checksum.

File	MD5
BNC ₂₀₁₄ CSV	1c041dbcb635855f3d8f4be9f3e1fced
DAIS item CSV	46af310f1633b8784f897e4482faed84
DAIS cleaned-judgement ZIP	e7b3ac6f2dd8e85bc93e6e20fc0f6ec1

Derived files and the source-verification note are stored in the repository; the README gives their paths. The principal reproduction commands are:

```
Rscript analysis/08_dais_acceptability_bridge.R
Rscript analysis/09_bnc2014_metadata_scope.R
Rscript analysis/10_bnc2014_paired_transport_cv.R
python3 analysis/06_denominator_and_figure_candidates.py
make
```

The cross-validation and bootstrap seed is 20260625. Session information is written to a derived text file. Raw DAIS files aren't committed.

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